

Highlights

- Imputation, feature selection and stacking were all fitted inside validation.
- Stacking matched but did not beat logistic regression (CV AUC 0.843 vs 0.842).
- RFECV cut inputs by 25% and kept glucose, BMI and family-history terms.
- Class weighting overestimated risk; calibration must be reported.
- Sensitivity-targeted thresholds transferred from development to hold-out data.